Time-aware Dimension Reduction for Exploring Trends in Literature

Brian D. Ondov, PhD[1], Xueqing Peng, PhD[1], Huan He, PhD[1], Hua Xu, PhD[1]
[1]Yale School of Medicine, New Haven, CT, USA

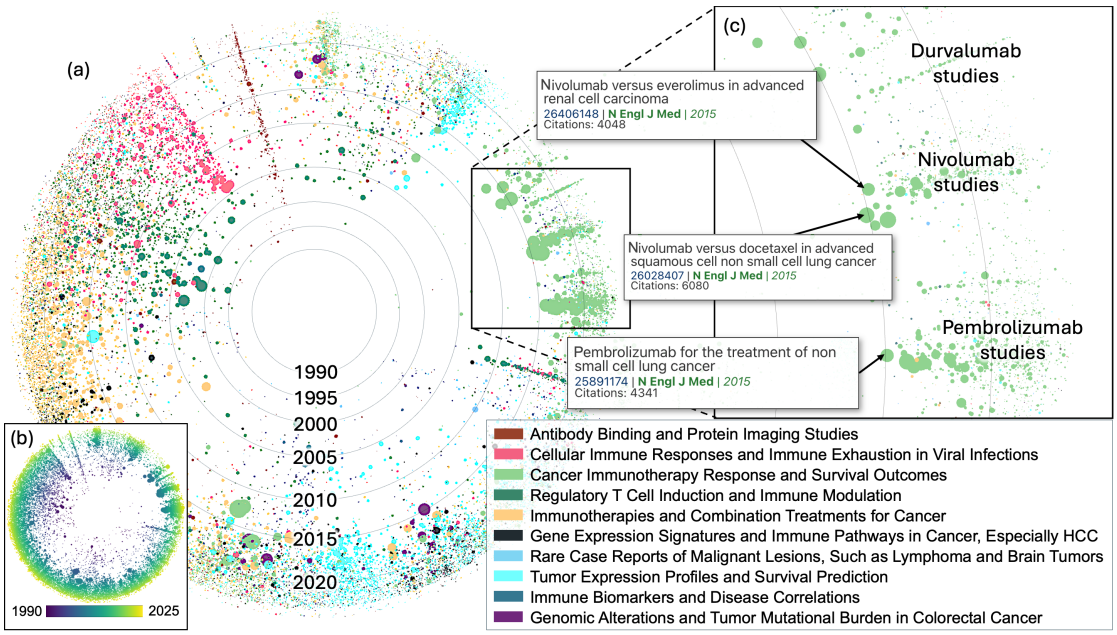

Figure 1. Time-aware dimension reduction for a collection of articles related to PD-1/PD-L1. Each point represents an article, with size corresponding to number of citations and distance from the center encoding date. In (a) color corresponds to topic, with labels summarizing articles in each topic in the legend. In (b), coloring articles by publication date shows the perfect temporal coherence induced by our new algorithm. In (c), our method shows highly cited studies on immune checkpoint inhibitors giving rise to narrowly focused lines of research on individual drugs.

Introduction: The million-plus papers indexed in PubMed yearly make methods for organizing and surfacing research more important than ever. LLM-based text embedding models[1], combined with manifold learning algorithms such as t-SNE[2], have allowed large collections of literature to be explored as 2-dimensional semantic "maps"[3]. However, these maps ignore dates of publication, a crucial part of understanding how lines of research arise and progress. **Methods:** To address this gap, we propose polar-coordinate reparameterization of the manifold learning objective: whereas t-SNE and similar algorithms optimize both x and y of each point, we directly encode time using radius, letting points move along the angular dimension to group with similar ones. **Results:** Our method creates plots with a clear time component while retaining the structure-preserving properties of t-SNE (Fig. 1). As a case study, we create a focused corpus of 26,342 PubMed articles related to PD-1 and PD-L1 and embed concatenations of paper titles and abstracts using LLM-embedder[1]. We then reduce from 768 dimensions to two with our new algorithm. The resulting layout shows clusters that align well with topics automatically assigned by Latent Dirichlet Allocation (Fig. 1a), but also reveal trends that would not be visible with standard t-SNE, such as key papers lying at the inception of fruitful lines of research in specific subdomains (Fig. 1c). Embedding quality is high by standard measures[4] of macroscopic structure (pairwise distance rank correlations) and mesoscopic structure (k-nearest class neighbors), though lower than t-SNE for microscopic structure (k-nearest neighbors). **Conclusions:** Our results demonstrate how time-aware dimension reduction is a valuable tool with broad implications for exploring large corpora of literature. Ongoing work includes development of a public Python package for wide availability and reproducibility.

References

1. Zhang P, Xiao S, Liu Z, Dou Z, Nie JY. Retrieve anything to augment large language models. arXiv preprint arXiv:2310.07554. 2023 Oct 11.
2. Van der Maaten L, Hinton G. Visualizing data using t-SNE. Journal of machine learning research. 2008 Nov 1;9(11).
3. González-Márquez R, Schmidt L, Schmidt BM, Berens P, Kobak D. The landscape of biomedical research. Patterns. 2024 Jun 14;5(6).
4. Kobak D, Berens P. The art of using t-SNE for single-cell transcriptomics. Nature communications. 2019 Nov 28;10(1):5416.